

Comprehensive—Volume + Area + Fraction

Cross-Topic integration (T1 · T5 · T9) · 65 minutes · 1-to-3 Online Lesson

Corresponding textbooks: "New Thinking in Primary School Mathematics (Second Edition)" Volume 5B Units 7-8 + Modern Education 5B Unit 18-20

Core traps: ● High Frequency

Sub-test cross-topic comprehensive questions, accounting for about 18-22% of Paper 1

Prerequisite knowledge: Class 21 (Circle + Section + Origami) · Class 22-23 (Fundamentals of Volume + Application) · Class 24-26 (Multiplication and Division of Fractions) · Class 27 (Division of Decimals)

Objectives of this class: ❶ Comprehensive application of volume and area formulas ❷ Application of fractions in volume ❸ Identification and avoidance of cross-topic traps

Our goals: ❶ Comprehensive application of volume and area formulas ❷ Application of fractions in volume ❸ Identification and avoidance of cross-topic traps

Student Name:

Class:

Date:

Time Spent:

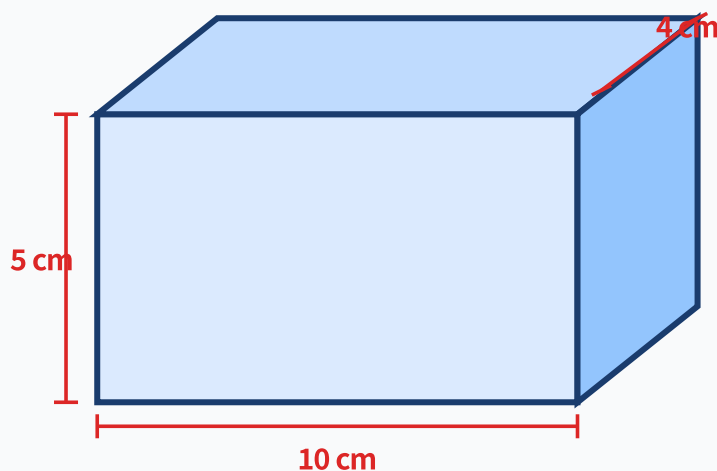
I.Warm-Up Questions (total 5 question , 5 minutes)

#	Question	Difficulty	Working Space (Show full working)
1	The cuboid is 6 cm long, 4 cm wide and 5 cm high. Volume = ?	Basic	
2	Triangular base 8 cm, height 6 cm. Area = ?	Basic	
3	The parallelogram has a base of 12 cm and a height of 5 cm. Area = ?	Basic	
4	The base area of a cuboid is 24 cm^2 and the height is 3 cm. Volume = ?	Advanced	
5	The volume of a cuboid is 120 cm^3 , its length is 6 cm, and its width is 5 cm. Seek high.	Advanced	

II.Core Knowledge + Worked Examples

Knowledge point 1: Comprehensive application of volume + area formula ● SSPA

- ① **Volume formula (cuboid/cube):** $V = \text{length} \times \text{width} \times \text{height}$; $V = \text{base area} \times \text{height}$
- ② **Area formula review:** Triangle $A = (\text{base} \times \text{height}) \div 2$ • Parallelogram $A = \text{base} \times \text{height}$ • Trapezoid $A = (\text{upper base} + \text{lower base}) \times \text{height} \div 2$
- ③ **Key distinction:** The area is the size of the "surface" (cm^2/m^2), and the volume is the size of the "space" (cm^3/m^3)
- ④ **Composite graphics strategy:** Divide into basic graphics $\rightarrow$ Find area/volume separately $\rightarrow$ Add or subtract
- ⑤ **Unit trap:** The base and height units must be consistent; the volume unit is cubic (3), not square (2)!



$$V = \text{length} \times \text{width} \times \text{height} = 10 \times 4 \times 5 = 200 \text{ cm}^3$$

□ **Example of trap detonation (the most important demonstration in this class)**

A rectangular container has a trapezoidal base (upper base 4 cm, lower base 6 cm, height 5 cm), and the height of the container is 10 cm. Find the volume of the container.

✗ **Common mistakes (70% students)**

$$V = (4+6) \times 5 \times 10 = 500 \text{ cm}^3$$

Forget about the area of a trapezoid ÷ 2! First find the base area = $(4+6) \times 5 \div 2 = 25 \text{ cm}^2$, then $\times 10 = 250 \text{ cm}^3$

✓ **Correct solution**

$$V = 250 \text{ cm}^3$$

① Base area = $(4+6) \times 5 \div 2 = 25 \text{ cm}^2$ ② $V = 25 \times 10 = 250 \text{ cm}^3$

🧠 **Tip: "First determine the surface and determine the body, and distinguish the basis of the formula; divide the area of a trapezoid by two, and then multiply the volume by the height; the unit is cubed, and the area is raised to the second power."**

⚠ **The most frequent error: Forget the area of the trapezoid ÷ 2, use (upper base + lower base) × height directly as the area. Remember that the trapezoidal formula must be divided by 2!**

⚠ **The second most common error: the unit of volume is written as cm^2 instead of cm^3 - area is square and volume is cubic!**

Knowledge Point - Worked Examples (write out complete steps: ①findbasearea ②findvolume ③mark unit)

#	Question	Difficulty	Working Space
Example 1	The side length of a cube is 5 cm. Find: (a) the area of a face (b) the volume		
Example 2	The cuboid is 8 cm long, 6 cm wide, and 4 cm high. Find the area and volume of the base.		
6	A rectangular parallelepiped with a square base (side length 7 cm) and height 10 cm. Find the volume.		
7	A rectangular container has a parallelogram base (base 8 cm, height 5 cm) and the height of the container is 12 cm. Find the volume.		
8	A rectangular parallelepiped has a base area of 36 cm^2 and a height of 9 cm. If the height is increased by 3 cm, how much will the volume increase?		

9	The picture below consists of two cuboids (stacked one above the other). Upper layer: $3 \times 3 \times 4$ cm; lower layer: $5 \times 5 \times 6$ cm. Find the total volume. Tip: Calculate separately and then sum.		
10	The side length of one cube is 2 times the length of the other cube. How many times the volume of the large cube is the volume of the small cube?		

Knowledge Point 2: Application of Fractions in Volume 🟡 SSPA Required Exam

Four ways to consider fractions in volume:

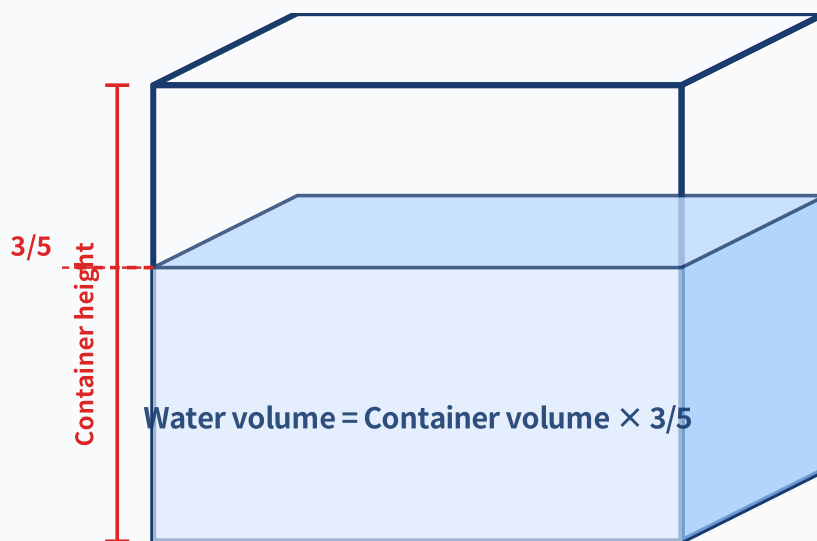
① **Side length contains fractions:** Length = $2\frac{1}{2}$ cm $\rightarrow$ Convert to false fraction and multiply: $V = \frac{5}{2}$

Container volume

② **Fraction of volume:** The container is packed $\frac{3}{4}$ of water $\rightarrow$ volume of water = volume of container $\times \frac{3}{4}$

③ **Find the volume of the parts and find the whole:** known $\frac{2}{5}$ The volume is $40 \text{ cm}^3 \rightarrow$ whole = $40 \div \frac{2}{5} = 100 \text{ cm}^3$

④ **Scaling:** Length enlargement $1\frac{1}{2}$ times $\rightarrow$ volume enlargement $(\frac{3}{2})^3 = \frac{27}{8}$ times



①

Side length includes fraction

Mixed fractions $\rightarrow$ improper

②

Find the fraction

Volume $\times$ Fraction = Partial Volume

③

Rounding the known parts

Part $\div$ Fraction = Whole Volume

④

scaling

Side length $\times$ k Volume $\times k^3$

fractions and then multiply to find the volume

example

Example 3: A cuboid is $2\frac{1}{2}$ cm long, 4 cm wide, and 3 cm high. Find the volume.

example

Example 4: A water tank has a volume of 80 L and is filled with $\frac{3}{5}$ water. What is the volume of water in L?

✗ Common mistakes

$$2\frac{1}{2} \times 4 \times 3 = 2.5 \times 12 = 30$$

After converting mixed numbers to improper fractions, the numerator and denominator are processed separately:
 $52 \times 4 \times 3 = 30$ ✓ (This question happens to be correct), but it is easy to get confused in complicated situations!

✓ Correct approach

$$V = \frac{5}{2} \times 4 \times 3 = 30 \text{ cm}^3$$

$$\textcircled{1} 2\frac{1}{2} = \frac{5}{2} \quad \textcircled{2} \frac{5}{2} \times 4 \times 3 = \frac{60}{2} = 30$$

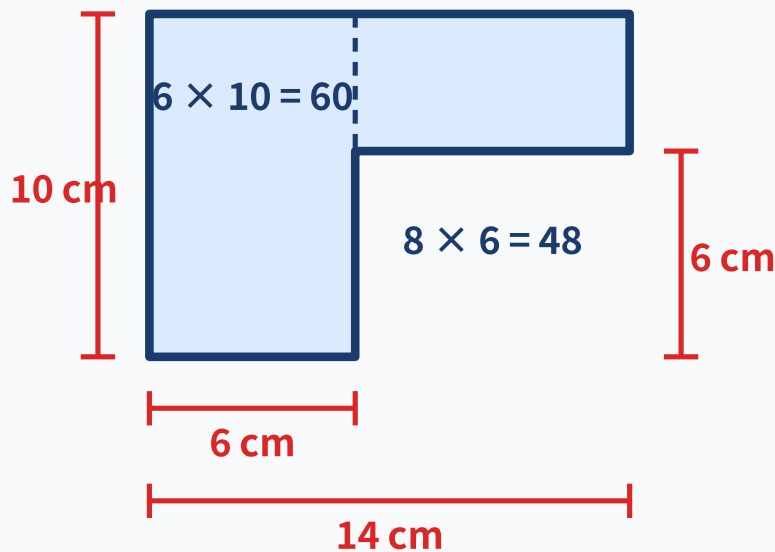
Knowledge Point 2 Synchronization practice

#	Question	Difficulty	Working Space
11	The side length of the cube is $1\frac{1}{2}$ cm. Find the volume. (Answer in fractions)		
12	A water tank has a volume of 60 L and contains $\frac{2}{3}$ water. How many liters of water is there?		
13	A fish tank $\frac{3}{4}$ has 45 L of water. What is the volume of the entire fish tank in L?		
14	The length of the cuboid is $\frac{3}{2}$ m, the width $\frac{2}{3}$ m, and the height is 2 m. Find the volume. (Answer with m^3)		
15	After the side length of the cube is enlarged by $1\frac{1}{3}$ times, how many times is the new volume the original? (Answer in fractions)		

Knowledge Point 3: Comprehensive Application—Volume + Area + Fraction Three-in-One  **SSPA Advanced**

Strategies for solving cross-topic comprehensive questions (three-step method for reading questions):

- ① **Circle all the numbers and units in the question**- Clarify which are areas, which are volumes, and which are fractions
- ② **Judge the order of operations**- First find the area (base area) → then find the volume → finally deal with the fraction ratio
- ③ **Check the consistency of the units**—m and cm cannot be multiplied directly! Convert to the same unit first



$$\text{Total area} = 60 + 48 = 108 \text{ cm}^2$$

example

Example 5: A rectangular water tank with a rectangular base (1 m long, 50 cm wide) and 80 cm high. The water tank contains $\frac{3}{5}$ water. Find the volume of water (answer in L, 1 L = 1000 cm³).

example

Example 6: The side length of cube A is 6 cm, and the side length of cube B is $\frac{2}{3}$ of A. What fraction of the volume of B is A?

 **Tip: "Unify the units first, then the area and then the volume; multiply the fractions last, and avoid all traps."**

 **High frequency trap: m and cm are mixed up - 1 m = 100 cm, but 1 m² = 10000 cm², 1 m³ = 1,000,000 cm³!**

Knowledge Point 3 Synchronization practice

#	Question	Difficulty	Working Space
16	The base area of a cuboid is 25 cm^2 and the height is 8 cm. What is the volume $\frac{3}{4}$ in cm^3 ?		
17	A fish tank is 50 cm long, 40 cm wide, and 30 cm high. (a) Find the volume of the fish tank (b) If water is filled to the height of SEP , what is the volume of water in L? $\frac{4}{5}$ Height, what is the volume of water in L?		
18	The side length of a cube water tank is 20 cm. Existing water 4000 cm^3 . What fraction of the tank's volume does water occupy? (Answer with the simplest fraction)		
19	The cuboid is 1.5 m long, 0.8 m wide, and 0.6 m high. (a) Volume = ? m^3 (b) If the space of $\frac{5}{8}$ has been used, how much m^3 is the remaining space?		
20	A cylinder has a base area of 78.5 cm^2 and a height of 10 cm. If the height is reduced by SEP , what is the new volume? (Hint: new high = $10 \times (1 - \frac{1}{5})$, what is the new volume? (Hint: new high = $10 \times (1 - \frac{1}{5})$))		

III. Lesson Layered Synchronization Practice

Basic layer (total 5 questions, everyone must do)

#	Question	Difficulty	Working Space
21	The cuboid is 10 cm long, 5 cm wide, and 4 cm high. Find the volume.		
22	The side length of the cube is 8 cm. Find the volume.		
23	Triangular base 12 cm, height 9 cm. Find the area.		
24	The upper base of the trapezoid is 4 cm, the lower base is 8 cm, and the height is 5 cm. Find the area.		
25	The volume of a cuboid is 72 cm^3 and the base area is 12 cm^2 . Seek high.		

Advanced layer (total 5 questions,   choose do)

#	Question	Difficulty	Working Space
26	A cuboid has a square base (sides 6 cm) and a height 15 cm. Find the volume.		
27	Water tank volume 120 L, used SEP . How much L space is left? $\frac{5}{6}$. How much L space is left?		
28	The side length of the cube is $1\frac{1}{3}$ cm. Find the volume. (Answer with improper fractions)		
29	Cuboid A: length 6 cm, width 4 cm, height 5 cm. Cuboid B: All sides are twice as long as A. How many times the volume of B is A?		
30	A combined solid: the lower cuboid is $10 \times 8 \times 5$ cm, and the upper cube with side length 4 cm is placed in the center. Find the total volume.		

 challenge layer (total 3 questions,  choose do, SSPAKiller Questions)

#	Question	Difficulty	Working Space
31	A rectangular container has an interior length of 30 cm, a width of 20 cm, and a height of 15 cm. After placing a stone, the water level rises by 2 cm. What is the volume of the stone in cm^3 ? Tips: Drainage method - stone volume = bottom area $\times$ water level rise height		
32	The side length of the cube is $\frac{3}{4}$ m. Find (a) the area of a surface (answer in m^2) (b) the volume (answer in m^3) (c) how many times the volume is the area of the base?		
33	A trapezoidal cylinder: the base is trapezoidal (upper base 5 cm, lower base 9 cm, height 4 cm), and the height of the cylinder is 10 cm. Find the volume. Tips: $V = \text{base area} \times \text{height} = \text{trapezoid area} \times \text{cylinder height}$		

IV.applicationquestionspecial topic training

#	Question	Difficulty	Working Space
34	A rectangular pool is 8 m long, 5 m wide, and 2 m deep. (a) What is the volume of the pool in m^3 ? (b) If the water depth is only SEP , what is the volume of water in m^3 ? $\frac{3}{4}$, what is the volume of water m^3 ?		

35	A rectangular container is 12 m long, 2.5 m wide, and 2.8 m high. (a) What is the volume of the container in m^3 ? (b) The loaded goods occupy SEP , how many m^3 of goods can be loaded in the remaining space? $\frac{5}{7}$, how many m^3 of goods can be loaded in the remaining space?		
36	A cube water tank has a side length of 40 cm. 8000 cm^3 of water is injected every minute. (a) What is the total volume of the tank in cm^3 ? (b) How many minutes does it take to fill the water tank? (c) After 5 minutes, what fraction of the water will be in the tank?		
37	A rectangular fish tank is 60 cm long, 30 cm wide, and 40 cm high. After first pouring 36 L of water and adding some stones, the water level rose by 5 cm. (a) 36 L What is the height of the water in the tank in cm? (1 L = 1000 cm^3) (b) What is the total volume of the stone in cm^3 ? (c) What is the final water depth in cm?		

V. Ultimate challenge area

#	Question	Difficulty	Working Space
 1	The length of a cuboid is 2 times the width, and the breadth is $1\frac{1}{2}$ times the height. If the height is 4 cm, find: (a) What is the width? (b) What is the length? (c) What is the volume? (d) What is the surface area? (Surface area = $2 \times (\text{length} \times \text{width} + \text{length} \times \text{height} + \text{width} \times \text{height})$)		
 2	A cube has a side length of 10 cm. A small cube with a side length of 2 cm is dug out in the center of each of its six faces (digging through to the opposite side). Find the volume of the remaining solid. Tips: Note that the perforations in the three directions will overlap, be careful not to deduct repeatedly!		
 3	A trapezoidal cylindrical pool: the bottom is trapezoidal (upper bottom 3 m, lower bottom 7 m, height 4 m), and the pool depth is 2 m. 32 m^3 of water available. (a) What is the total volume of the pool in m^3 ? (b) What fraction of the total pool is the existing water? (Answer in simplest fractions) (c) How many m^3 of water must be injected to fill it completely?		

Cross-topic integration: volume + area + fraction, a universal problem-solving framework essential for score examinations

When encountering comprehensive questions, deal with them in the following order:

- ① **Read the questions:** Circle all the numbers, mark the units (cm/m/L/cm³/m³), and find out "a fraction"
- ② **Unify the units:** Unify all lengths into the same unit (all cm or all m)
- ③ **Area first, then volume:** If you need the base area → use the corresponding formula (rectangle/triangle/trapezoid/parallelogram)
- ④ **Processing fractions:** Volume × fraction = volume of part; part ÷ fraction = whole
- ⑤ **Check the answer:** Are the units correct? Are fractions reduced? Is the answer reasonable?

VI. Class after homework

Basic must-do questions (total 5 questions, must Show full working)

#	Question	Difficulty	Working Space
H1	The cuboid is 7 cm long, 5 cm wide, and 3 cm high. Find the volume.		
H2	The side length of the cube is 9 cm. Find the volume.		
H3	A cuboid has a base area of 28 cm ² and a height of 6 cm. Find the volume.		
H4	Water tank volume 90 L. Installed $\frac{2}{5}$ water. How many liters of water is there?		
H5	A cuboid is 1.5 m long, 0.8 m wide, and 0.5 m high. Find the volume (answer in m ³).		

Advanced choose do question (total 3 questions,  choose do)

#	Question	Difficulty	Working Space
H6	The cuboid is $2\frac{1}{2}$ cm long, 4 cm wide and 3 cm high. Find the volume.		
H7	A cube water tank has a side length of 25 cm. Pour 8 L of water first, then add the stones, and the water level rises by 3 cm. What is the volume of the stone in cm ³ ? (1 L = 1000 cm ³)		
H8	Cuboid A: 6×4×5 cm. Cuboid B: All side lengths are $1\frac{1}{2}$ times that of A. How many times the volume of B is A? (Answer in fractions)		

VII. The Lesson core common errors summary

✅ Self-examination in this hall (tick after completion)

☐ I know the pitfalls of solving each knowledge point

☐ I can complete 🍌 basic questions independently

☐ I can challenge 🌿 advanced questions

☐ I remember the formula

🎯 Review of Learning Objectives - After completing this lesson you should be able to:

☐ Identify all trap types in our hall

☐ Solve 🍌 basic questions independently (100% correct)

☐ Challenge 🌿 Advanced questions (80%+ correct)

☐ Explain the lesson formula to classmates

#	common error	Correct Approach
1	Forget the area of the trapezoid $\div 2$ (upper base + lower base) $\times$ height, not the area of the trapezoid!	Trapezoidal area = (upper base + lower base) $\times$ height $\div 2$, you must also divide 2 when using it to calculate the base area.
2	The volume unit is written incorrectly SEP : cm^3 is written as cm^2 : cm^3 is written as cm^2	Area $\rightarrow$ square (2), volume $\rightarrow$ cubic (3). Check the index before each answer
3	m/cm conversion error SEP : $1\text{m}=100\text{cm}$ but $1\text{m}^3 \neq 100\text{cm}^3$: $1\text{m}=100\text{cm}$ but $1\text{m}^3 \neq 100\text{cm}^3$	$1\text{m}^3 = 100 \times 100 \times 100 = 1,000,000\text{cm}^3$. Three-dimensional conversion requires conversion in each dimension
4	The fractions are not reduced after multiplication : $\frac{6}{12}$ Take it directly as the answer	The last step is to check whether the numerator and denominator have common factors, which must be reduced to the simplest
5	"Part $\div$ Fraction = Whole" is reversed SEP : Use multiplication instead of division : Use multiplication instead of division	Given that $\frac{3}{5}$ is 60 $\rightarrow$ overall = $60 \div \frac{3}{5} = 100$ (not $60 \times \frac{3}{5}$)
6	The side length is enlarged k times $\rightarrow$ the volume is enlarged k times (Wrong!)	The side length is enlarged k times $\rightarrow$ the volume is enlarged k^3 times (cubic!)
7	Misunderstanding of drainage method SEP : Thinking that the rising height of the water level = the height of the stone : Thinking that the rising height of the water level = the height of the stone	The volume of the stone = the bottom area of the container $\times$ the rising height of the water level (not the height of the stone)

Lam Fung Academy • LF Academy • We don't teach math. We teach trap avoidance.

📖 Related topics: L03 Area of parallelograms and triangles • L04 Area of trapezoidal polygon • L05 Area trap special project •
Related topics: L25 Volume word problems • L26 Volume concept • L27 Composite three-dimensional drainage method